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ARTICLE

Causal-Role-Aware Extractive Evidence Selection for Power Grid Reliability Reports

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ABSTRACT: Power-grid maintenance and reliability documentation contains causal event narratives whose evidence is buried across dense technical prose, which makes post-event review difficult when analysts must trace claims about causes, impacts, protection behavior, and corrective actions back to specific report sentences. Prior extractive summarization and retrieval systems identify salient or topically related sentences, but they are not designed to return sentence identifiers that answer role-conditioned causal questions over North American Electric Reliability Corporation (NERC) reports or report excerpts. This paper formulates the problem as causal-role-aware evidence selection and introduces C^{2}GES, a lightweight deterministic reranker that combines query relevance, causal-role compatibility, and role-selective causal-chain consistency; the "counterfactual" scope is limited to ablation-style component removal, not physical intervention or structural causal estimation. The method is evaluated on a self-built benchmark of 40 public NERC reports or report excerpts with 200 role-conditioned questions covering five causal roles and 608 agent-verified candidate evidence sentence identifiers. On this benchmark, the full C^{2}GES reranker improves evidence F1 by approximately 41% relative over TF-IDF query retrieval, approximately 31% over Okapi BM25 query retrieval, and approximately 51% over SBERT semantic query retrieval, and paired document-cluster bootstrap comparisons support the primary gain over lexical query retrieval (mean difference +0.0861 evidence F1; 95% confidence interval $[0.0581, 0.1145]$; $p < 0.001$). In absolute terms, the full reranker reaches 0.2983 evidence F1 under a strict top-3 sentence-identifier exact-match protocol, an operating range consistent with the difficulty of five-way role-conditioned selection over long technical reports, where the strongest generic baselines remain below 0.23. Protocol-matched learned and LLM baselines bound the method's position: C^{2}GES stays ahead of two neural rerankers on point estimates in its transparent, CPU-deployable class (0.2604 and 0.2787 evidence F1; one comparison borderline-significant, one statistically indistinguishable), while a zero-shot LLM selector reaches 0.5887, an honest upper reference that offers no per-term score decomposition and requires a remote proprietary endpoint. Ablations show that role compatibility supplies the main improvement, while role-selective graph consistency provides a small auxiliary gain concentrated in the propagation/response and mitigation roles. These results support C^{2}GES as a scoped, auditable ranking aid for causal-role evidence selection in public NERC report analysis using agent-verified candidate labels.

KEYWORDS: Extractive summarization; causal evidence selection; power grid maintenance reports; NERC reliability reports; causal-role reranking; evidence retrieval

1 Introduction

Power-grid maintenance and reliability reports preserve operational knowledge about disturbances, protection actions, equipment behavior, remediation, and lessons learned. In this paper, the phrase "maintenance reports" is used broadly to refer to public NERC reliability, disturbance, lessons-learned, and event-focused report excerpts that document grid events and corrective actions. Their value lies not only in the events they describe, but in the causal mechanisms that connect a triggering condition to a downstream operational impact and an eventual mitigation. Such reports are used in post-event review, where traceable sentences can support claims about causes, impacts, protection behavior, operator response, and corrective actions. This requirement is sharpened by the growing role of data-driven analysis in electric power systems, where AI methods are increasingly discussed for grid modeling, reliability, forecasting, and digital infrastructure, but where textual evidence from reports remains difficult to operationalize [1–4]. C^{2}GES studies this problem as evidence-grounded causal-role sentence selection: given NERC report sentences and a causal role question, the system returns evidence sentence IDs. This formulation follows the spirit of query-focused summarization benchmarks such as QMSum, where the information need conditions the relevant content, but adapts the unit of prediction to sentence-level causal evidence rather than a fluent generated answer [5]. Building on this observation, the paper treats selected evidence as the primary model output and treats any downstream causal summary as a packaging layer over that selected evidence.

Generic extractive summarization and generic retrieval leave a gap for this setting because relevance alone does not determine causal role. A sentence describing load loss may be central to an event narrative, but it may answer an impact question rather than a root-cause question; similarly, a sentence mentioning relay operation may describe propagation, mitigation, or both depending on its local context. Classical and modern extractive summarization methods rank sentences by salience, centrality, redundancy control, or learned sentence matching [6–8]. Graph-based summarizers extend this idea by modeling sentence or heterogeneous document structure, including TextRank-style ranking, heterogeneous graph neural networks, and multiplex sentence graphs [9–12]. Query-focused systems instead condition selection or generation on an explicit information need, and recent work has examined query-focused meeting summarization, salient-content ranking, and evidence attribution for long-context summarization [13–16]. In contrast to these broader summarization settings, NERC report analysis requires distinguishing among causal roles that are topically close but operationally different. The benchmark therefore asks whether role-conditioned evidence selection improves over lexical query retrieval, semantic query retrieval, positional baselines, and graph-centrality summarization baselines on the same sentence pools.

Causal language further complicates the task because causal terms in text are suggestive but insufficient for operational event reconstruction. Prior work in causal NLP has emphasized that extracting, representing, or reasoning about causal relations requires care about what is being estimated or inferred [17–20]. Other work has studied explainable causal reasoning datasets, procedural causal event reasoning, document-level causal relation extraction, and causal structure induction for summarization [21–24]. These directions motivate causal-role awareness, but the present task is narrower: it evaluates whether a reranker can recover annotated evidence sentences for role-specific questions in public power-grid reports or report excerpts. The "counterfactual" terminology in C^{2}GES is correspondingly scoped to ablation-style component removal. By removing scoring components, the experiments ask what changes when role compatibility or chain consistency is absent, rather than simulating alternative physical grid trajectories or estimating structural causal effects. This scope matters because power-system disturbances involve protection settings, operator actions, equipment states, and regional grid behavior that are not fully recoverable from report prose alone.

C^{2}GES is a lightweight reranking method for this role-conditioned evidence task. For each document sentence, question, and causal role, C^{2}GES combines a query relevance score with a role compatibility score and a local causal-chain consistency score. Query relevance supplies the retrieval backbone, role compatibility biases the ranker toward sentences that express the requested causal function, and chain consistency is activated only for propagation/response and mitigation questions, where nearby causal-mechanism context is most likely to affect the selected evidence. Fig. 1 gives an overview of the architecture. The method is implemented as a sentence ranker over the local benchmark at `verification_pilot/agent_audit_40doc/`, with the post-freeze revision package under `post_freeze_revisions/role_selective_graph_gate/`; the Executor revision records completed metrics in the run artifact used for this revised manuscript. The empirical pattern is clear on the candidate benchmark: the full method outperforms lexical and semantic query retrieval, and the role ablation identifies causal-role compatibility as the dominant source of the improvement. The graph term remains auxiliary but measurable in the role-selective revision, which turns the main technical message toward scoped role-aware reranking with a small structural adjustment rather than broad graph inference.

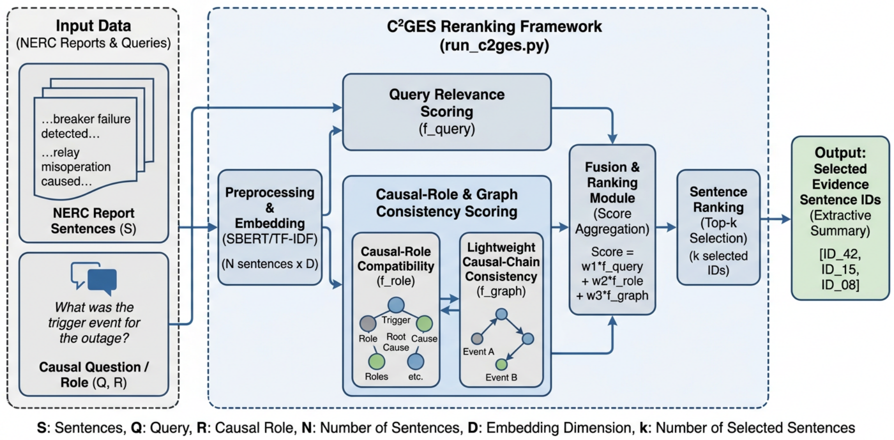

Figure 1: Overview of the C^{2}GES architecture. For each report sentence, question, and causal role, the reranker combines query relevance, role compatibility, and a role-selective causal-chain consistency signal into an additive score and emits the top-K evidence sentence IDs.

This paper makes three contributions:

- It defines an evidence-grounded causal-role selection benchmark over public NERC power-grid reports and excerpts, with role-conditioned questions and sentence-ID candidate evidence labels covering trigger events, root causes, propagation or response, impacts, and mitigation.
- It introduces C^{2}GES as a lightweight causal-role-aware reranker for the task of mapping NERC report sentences and a causal-role question to evidence sentence IDs, with role-selective graph consistency framed as an auxiliary chain-consistency feature rather than as full graph neural inference or power-system simulation.
- It evaluates C^{2}GES against a controlled set of extractive and retrieval baselines and reports ablations showing that role compatibility drives the main improvement, while role-selective causal-chain consistency provides a small auxiliary gain concentrated in propagation/response and mitigation roles.

The rest of this paper is organized as follows. Section 2 situates the work in query-focused summarization, extractive graph ranking, causal NLP, evidence retrieval, and power-system reliability analysis. Section 3 formalizes the task and benchmark, including the role-conditioned evidence schema and label provenance. Section 4 presents C^{2}GES as a technical reranking model, including a methods-integrity discussion of cue provenance. Section 5 describes the experimental protocol, reports quantitative results, ablations, and qualitative analysis, and discusses the findings. Section 6 concludes with limitations and future work.

2 Related Work

2.1 Query-Focused and Extractive Summarization

Query-focused summarization provides the closest task family because it conditions content selection on an explicit information need rather than a document-global notion of salience. QMSum introduced query-based multi-domain meeting summarization and made the query a central part of the summarization objective [5]. Subsequent work has explored neural query-focused summarization, query-relevant knowledge, query-utterance attention, and utterance ranking for meeting summaries [13,14,25,26]. Recent studies also examine contrastive salient content, learned ranking, and evidence attribution for long-context query-focused summarization [15,16,27]. These studies show that query conditioning can change which information should be selected, but the evaluated output is commonly a natural-language summary or an answer-oriented content set. C^2GES differs by making sentence-ID evidence recovery the evaluated output and by conditioning the query on causal roles in NERC reports or report excerpts, which forces the system to distinguish evidence for triggers, causes, propagation or response, impacts, and mitigation.

Extractive summarization has long used sentence ranking, centrality, redundancy control, and text matching to identify salient content. Extractive summarization as text matching provides a direct framing for sentence-level selection [6], while adaptive redundancy-aware ranking and rank-fusion methods show how salience signals can be combined without abstractive generation [7,8]. Work on TextRank variants and centrality-based sentence selection illustrates the strengths and limits of graph-style salience when the information need is generic [9,28]. Studies of BERT-based extractive summarization further reinforce that extractive methods are attractive when traceability to source text matters [29]. C^2GES builds on this extractive tradition but changes the target: the correct sentence is not simply the most central sentence in a report, but the sentence that supports a specified causal role for a specified question.

2.2 Graph-Based Document Modeling

Graph-based summarization methods motivate the structural component of C^2GES, although the present method uses a lightweight consistency signal rather than a full graph neural model. Heterogeneous graph neural networks represent documents with multiple node and edge types, allowing sentence, word, and discourse information to interact during extraction [10]. Multiplex graph neural summarization and topic-aware graph neural networks similarly show that graph structure can improve sentence selection when relations among textual units are informative [11,12]. Other work has explored bipartite graph pre-training, multi-granularity heterogeneous graphs, discourse-aware unsupervised summarization, and distance-augmented sentence graphs for long-document extraction [30–33]. In contrast to these architectures, C^2GES does not train a neural graph model; it uses causal-chain consistency as an auxiliary reranking feature so that the primary empirical question remains whether role-aware scoring improves evidence selection. This distinction is central because the benchmark has 40 reports or report excerpts, which is appropriate for transparent reranking and ablation analysis but not for broad claims about learned graph inference.

Graph centrality has a different failure mode in role-conditioned report analysis. TextRank-style and LexRank-style methods can identify sentences that are similar to many other sentences, yet role-specific evidence can appear in technical clauses that are not globally central. A mitigation sentence may occur near the end of an excerpt, an initiating equipment condition may appear once in a time-line passage, and an impact sentence may be marked by quantities rather than by repeated vocabulary. Prior graph summarizers therefore provide strong baselines for document structure, but they do not solve the role-disambiguation problem by themselves [9,28]. C^2GES uses graph information differently: instead of treating centrality as the main signal, it asks whether local causal-chain consistency can adjust a query and role score when the requested role is structurally contextual. The results later show that this graph term is auxiliary, with its measured effect concentrated in propagation/response and mitigation.

2.3 Evidence Retrieval and Causal NLP

Evidence retrieval research highlights the importance of returning support passages or sentences that justify downstream answers. Multi-step evidence retrieval for fake-news detection, alignment-based evidence retrieval for multi-hop question answering, and hybrid evidence retrieval all study variants of matching information needs to supporting text [34–36]. These studies are relevant because C^2GES evaluates evidence sentence IDs directly, but the evidence need is specialized to causal roles in power-grid reports. The role conditioning makes a false positive more subtle than topical irrelevance: a sentence may be about the same disturbance and still fail to answer the requested causal role. In contrast to evidence retrieval tasks where support may be any passage that entails an answer, the present benchmark requires the selected sentence IDs to align with candidate labels for a predefined causal function.

A rapidly growing line of work uses large language models directly as zero-shot rerankers. Pairwise ranking prompting shows that moderate-sized LLMs can rival supervised rankers on standard retrieval benchmarks [37], setwise prompting improves the efficiency of such zero-shot LLM ranking [38], and self-calibrated listwise reranking extends LLM rerankers with globally comparable relevance scores [39]. This paper evaluates representatives of both families as protocol-matched conditions: a cross-encoder reranker, a BGE-style reranker, and a zero-shot LLM selector are run with the released harness on the same benchmark protocol and reported in Section 5.5, and Section 5.7 discusses the transparency and cost tradeoff between the deterministic and learned/LLM families that these results quantify.

Causal NLP supplies the conceptual pressure behind role-aware evidence selection. Surveys and evaluations of causal inference and causal reasoning in NLP distinguish prediction, interpretation, causal relation extraction, and stronger forms of causal inference [17–20]. Dataset and modeling work on explainable causal reasoning, procedural causal events, document-level causal relation extraction, and causal structure for summarization show that causal information in text can be represented in multiple ways [21–24]. C^2GES uses this literature to motivate causal-role categories, but it evaluates retrieval of supporting report sentences rather than discovery of new causal structures. That distinction keeps the method aligned with auditable evidence selection and prevents over-reading report prose as a complete causal model of power-system dynamics.

2.4 Power-System Reliability Reports and Domain-Specific NLP

Power-grid reports are a demanding domain because technical incident narratives combine physical infrastructure, protection logic, operator response, reliability standards, and cascading effects. Work on cyber attacks and cascading failures, AI for power-grid systems, anomaly detection, and digital-grid opportunities shows the broader importance of causal and operational reasoning in grid analysis [1,4,40,41]. Studies of power-grid mathematical models, digital twins, grid fragility, and reliability contributions of different generation types further indicate that grid events must be interpreted in context rather than as isolated text snippets [2,42–44]. C^2GES contributes a complementary NLP benchmark for public NERC report evidence, focusing on traceable causal-role sentence selection instead of simulation, forecasting, or document-level classification.

Domain-specific NLP for power-grid reports must also respect the distinction between textual evidence and operational ground truth. A report sentence can support a claim about a relay trip, a generator step-up transformer fault, load shed, underfrequency behavior, restoration, or a corrective action plan, but the sentence alone may not contain all measurements or engineering context needed for complete event reconstruction. This is why C^{2}GES treats report text as evidence for role-conditioned questions rather than as a substitute for protection studies, power-flow simulation, or expert adjudication. In contrast to systems that aim to predict grid state from sensor or topology data, the present method ranks sentences in public documentation. The contribution is therefore a scoped bridge between evidence retrieval and power-system reliability analysis: it gives analysts a reproducible way to test whether causal-role cues improve retrieval of candidate evidence sentences in NERC report prose.

3 Task and Benchmark

The benchmark formalizes causal report analysis as sentence-ID evidence selection. Each example consists of a public NERC report or event-focused excerpt, a causal role, and a role-conditioned natural-language question. The prediction target is a set of sentence identifiers rather than a generated paragraph:

$$D_i, q_{i,r}, r \to \hat{Y}_{i,r}^{K}, \tag{1}$$

where D_i is the sentence-segmented report or excerpt, $q_{i,r}$ is the question for role r, and $\hat{Y}_{i,r}^{K}$ is the top-K predicted evidence sentence set. The benchmark uses $K = 3$ as a common prediction budget for all systems. This normalization is important because evidence sets vary in size, and because the task rewards a method for selecting concise support rather than returning long excerpts.

The local benchmark contains 40 NERC reports or report excerpts, 200 causal-role questions, and 608 candidate evidence sentence IDs. Each document contributes five questions, corresponding to trigger event, root cause, propagation or response, impact, and mitigation. These roles were chosen because they reflect the causal-mechanism structure used in disturbance and reliability analysis: an event begins, underlying causes and system responses shape its propagation, operational impacts follow, and mitigation or recommendations close the loop. The schema validation found 0 missing evidence IDs in the final dataset used for the Executor run, which allows the evaluation to use the provided sentence identifiers directly.

The benchmark was constructed in two recorded stages, and the process records are preserved in the supplement metadata rather than summarized away. The first 25 documents come from an earlier agent-audit release (`verification_pilot/agent_audit_25doc`); 15 additional processed NERC documents were then annotated with the same five-role schema and reviewed by a separate independent agent-verifier pass. For those 15 new documents, the recorded outcome is 6 clean passes, 8 passes with minor repairs, and 1 initial failure repaired by narrowing the answer to the cited evidence; the initial parallel verification attempt failed on 5 batches with API rate-limit (HTTP 429) errors and was rerun sequentially. All verifier-raised issues were repaired by narrowing unsupported wording, correcting evidence IDs, or reframing broad annual-report excerpts as event- or mechanism-focused excerpts. Schema validation over the final dataset reports 0 evidence-ID errors and exactly 40 questions per causal role, and the metadata flags one document (a 2016 State of Reliability excerpt) for suspiciously low evidence diversity.

These counts are stated here so that the provenance disclosure below is concrete and auditable rather than generic.

The label provenance is deliberately limited. The evidence labels are agent-rewritten and agent-verified candidate labels produced with large-language-model-based tooling, not human-gold or expert-gold annotations. The benchmark is therefore best understood as an auditable candidate benchmark for method development, not as a final adjudicated power-system evidence corpus. This limitation affects claim wording throughout the paper: reported scores measure agreement with candidate evidence annotations, and the paper does not claim expert-level extraction performance.

Some source documents are event-focused excerpts from broader reliability or lessons-learned reports. This design keeps each example focused enough for sentence-ID evaluation, but it also means that some operational context may be outside the local excerpt. The evaluation therefore asks whether a method can recover the labeled evidence available in the provided report text, not whether it can reconstruct all physical, operational, or regulatory causes of the underlying grid event. Building on this scope, the benchmark supports method comparison for role-aware evidence selection while leaving expert adjudication and full operational reconstruction to future work.

The benchmark artifacts are preserved for auditability. The dataset is stored under `verification_pilot/agent_audit_40doc/`. The post-freeze role-selective graph-gate package records the revised protocol under `post_freeze_revisions/role_selective_graph_gate/experiment_outputs/c2ges_role_selective_graph/`, including `cv_protocol.json` for document-level fold assignment and selected configurations, `heldout_predictions.jsonl` for held-out full and fixed-legacy C^{2}GES predictions, and `metadata.json` for command lines, dependency status, dataset paths, output paths, and label-provenance caveats. The completed metric source used for the quantitative tables is the valid completed run artifact from this package; null placeholder result files are not used as experimental evidence. A detailed mapping from each reported table to its evidencing artifact is given in the Supplementary Materials (Section S4).

4 Method

4.1 Problem Formulation and Scoring Objective

C^{2}GES models power-grid report analysis as role-conditioned evidence sentence reranking. Let a report or excerpt be a sequence of sentences $D_i = \{s_{i1}, \ldots, s_{in_i}\}$, and let $r \in \mathcal{R}$ denote one of five causal roles, where $\mathcal{R} = \{\text{trigger, root cause, propagation/response, impact, mitigation}\}$. For each document-role pair, the benchmark provides a natural-language role question $q_{i,r}$ and an agent-verified candidate evidence set $Y_{i,r} \subseteq \{1, \ldots, n_i\}$. The model predicts a ranked list of sentence indices and returns the top K IDs, $\hat{Y}^K_{i,r}$, with $K = 3$ in the experiments. The objective is operational rather than generative: the system is evaluated by how well $\hat{Y}^K_{i,r}$ matches $Y_{i,r}$, and any later prose summary is treated as a packaging layer over selected evidence.

C^{2}GES builds on the observation that the relevant unit in this task is not document salience, but causal-role support. Query-focused summarization has shown that explicit information needs change what content should be selected [5,13], and extractive summarization as sentence matching provides a natural basis for scoring sentence-question compatibility [6]. Graph-based extractive summarization further motivates structural scoring among document units [10,11], while causal NLP work motivates separating causal prediction and textual evidence from stronger causal inference claims [17]. Building on this observation, C^{2}GES uses a transparent additive reranking function whose terms correspond to query relevance, role compatibility, and lightweight causal-chain consistency.

For a candidate sentence s_{ij}, question $q_{i,r}$, role r, and document D_i, C^{2}GES computes a scalar score

$$\mathrm{Score}_\theta(s_{ij}, q_{i,r}, r, D_i) = w_q Q(s_{ij}, q_{i,r}) + w_r R(s_{ij}, r) + w_g \mathbb{I}[r \in \mathcal{R}_g] G(s_{ij}, D_i, r), \tag{2}$$

where $\theta = (w_q, w_r, w_g)$ are nonnegative scoring weights and $\mathcal{R}_g = \{\text{propagation/response}, \text{mitigation}\}$ is the set of graph-active roles. The term Q measures question-sentence relevance, R measures compatibility between the sentence and the requested causal role, and G measures whether the sentence fits a local causal-chain pattern in the document. For trigger, root-cause, and impact questions, the revised full scorer deliberately uses the same query-plus-role form as the no-graph variant because the post-freeze audit found no reliable benefit from forcing graph pressure into those roles. Sentences are sorted by Score_θ, ties are resolved by stable document order, and the top K sentence IDs are emitted as evidence. This design keeps the method auditable because each score can be decomposed into interpretable evidence for why a sentence was selected.

4.2 Query, Role, and Chain Components

The query relevance component $Q(s, q)$ supplies the retrieval backbone. It measures how closely a candidate sentence matches the role-conditioned question, using lexical overlap and normalized text-matching signals aligned with the TF-IDF query retrieval baseline. Formally, if $\phi(\cdot)$ maps text to a normalized sparse vector, the lexical relevance score can be written as

$$Q_{\mathrm{lex}}(s, q) = \frac{\phi(s)^\top \phi(q)}{\|\phi(s)\|_2 \|\phi(q)\|_2}. \tag{3}$$

When semantic sentence embeddings are available for comparison, the same form applies to dense embeddings $\psi(\cdot)$, giving $Q_{\mathrm{sem}}(s, q) = \cos(\psi(s), \psi(q))$. In C^{2}GES, the query component is the base signal rather than the full model, because the benchmark asks whether a sentence answers a specific causal-role question instead of merely sharing event vocabulary. The query-only ablation operationalizes this distinction by retaining Q while removing the causal-role and chain terms.

The causal-role compatibility term $R(s, r)$ is the central role-aware mechanism. It maps each sentence to a role-likelihood-style score using role-specific lexical, causal, and domain cues. A trigger sentence can name the initiating disturbance or operational event, with cues such as "fault," "trip," "loss of generation," "oscillation began," "line outage," or "relay operation." A root-cause sentence can express underlying equipment, protection, planning, or procedural causes, with power-system vocabulary such as "generator step-up transformer," "surge arrester," "protection setting," "failed breaker," "incorrect relay setting," or "equipment failure." A propagation or response sentence can describe cascading behavior, relay action, system response, or operator intervention, including "underfrequency," "voltage ride-through," "automatic generation control," "operator action," "load shed," or "intercept valves." An impact sentence can state load loss, outages, affected customers, reliability consequences, generation reduction, equipment damage, or net-load changes. A mitigation sentence can describe restoration, repair, corrective action plans, recommendations, protection-setting changes, equipment replacement, training, or procedural revisions. These role cues are not treated as causal proof; they are scoring features that make the ranker sensitive to the difference between topically similar but role-incompatible sentences.

A compact way to express this compatibility score is

$$R(s, r) = \sigma\left(\alpha_r^\top f_{\mathrm{role}}(s) + \beta_r^\top f_{\mathrm{domain}}(s) + \gamma_r^\top f_{\mathrm{cue}}(s)\right), \tag{4}$$

where f_{role} contains role-indicative terms, f_{domain} contains power-grid report vocabulary, f_{cue} contains causal and temporal markers, and σ normalizes the resulting score. In the implementation, this notation is not a learned neural parameterization. It is a deterministic cue-weight scheme: each matched cue contributes $1.0 + 0.15(\ell - 1)$, where ℓ is the number of words in the cue; numeric engineering units such as MW, GW, kV, Hz, or percent add 0.8; and domain abbreviations such as SLG, RAS, IBR, BESS, PV, ERCOT, CAISO, MISO, NERC, and WECC add 0.25. Scores are min-max normalized within the sentence pool for each document-role question. This choice is practical for the 40-document benchmark because it gives role-aware behavior without requiring a large supervised neural model, and it makes ablations directly interpretable.

The chain-consistency component $G(s, D, r)$ is a lightweight causal-chain consistency score. It uses local document context to reward sentences that are compatible with nearby role structure, such as propagation and response evidence occurring between initiating events and consequences, or mitigation evidence appearing near corrective-action and restoration language. The revised scorer gates this term by role: G is active for propagation/response and mitigation, and inactive for trigger, root-cause, and impact. This gate is a methodological constraint rather than a post hoc label edit; all conditions retain the same documents, labels, $K = 3$ prediction budget, metric, and document-level fold protocol. Let $p(j, k)$ be a proximity kernel between candidate sentence s_j and neighboring sentence s_k, and let $C(s_k, r')$ estimate whether s_k expresses a neighboring causal role r'. The consistency term can be written as

$$G(s_j, D, r) = \sum_{k \neq j} p(j, k) \sum_{r' \in \mathcal{R}} A_{r,r'} C(s_k, r'), \tag{5}$$

where A is a lightweight role-transition matrix encoding plausible causal-chain adjacency. For example, propagation evidence is more chain-compatible when nearby text also expresses initiating events or impacts, while mitigation evidence is more compatible when it appears near response or corrective-action language. This term is a reranking feature over report sentences, not a learned structural causal graph, a full graph neural network, or a power-system counterfactual simulator. Its purpose is to add weak document-structure pressure where local chain context is expected to matter, while preserving sentence-level evidence selection as the primary task.

4.3 Role Cue Interpretability

The role scorer uses interpretable power-system cues: event-start and trip language for trigger events; equipment or protection failures for root causes; grid, relay, operator, and control responses for propagation/response; measured MW, load, customer, generation, or damage consequences for impacts; and restoration, repair, recommendation, or prevention language for mitigation. The frozen implementation contains 15/21/21/19/19 cues for these five roles, plus engineering-unit and domain-abbreviation bonuses. Thus, unlike a lexical retriever, it can prefer a measured impact sentence over a trigger sentence that shares outage vocabulary. Complete cue examples, constants, and the deterministic scoring specification are retained in Supplementary Section S7.

4.4 Methods Integrity: Cue-Lexicon Provenance and Label–Cue Circularity

The manually curated 15/21/21/19/19 cue lexicons and fixed bonuses (Supplementary Section S7) were frozen before document-level 5-fold mixture-weight selection, but cue curation itself was not protected by a held-out split. Because the evidence labels are agent-rewritten and agent-verified rather than expert gold, cues and labels may share lexical preferences; the role gain may therefore partly measure agreement between annotation-style priors. The same caveat applies oppositely to the zero-shot LLM baseline because LLM agents produced and verified the labels. Reported scores should be read as agreement with candidate labels until the planned independently dual-annotated 50–100-question subset reports Cohen's κ and agent-label agreement.

The selected graph signal $G = \text{minmax}(R(s, r) \times \text{chain}(s, D, r))$ contains role compatibility. Thus, the no-graph ablation removes a role–structure interaction, not a purely structural signal; its small delta estimates only the marginal value of chain-modulated role evidence in the two graph-active roles. Supplementary Section S7 discloses the full formula and constants.

4.5 Ablation Scope and Algorithm

The ablation aspect of C^{2}GES is implemented through component removal rather than physical intervention modeling. The method asks how selected evidence changes when role compatibility is removed, when role-selective chain consistency is removed, or when only query relevance remains. In notation, the query-only variant sets $w_r = w_g = 0$, the no-role variant sets $w_r = 0$, and the no-graph variant sets $w_g = 0$. These contrasts give a direct empirical estimate of the marginal contribution of each scoring family under the same candidate set and prediction budget. This ablation view is important in a small technical benchmark because it ties the method claim to observable evidence-selection behavior without claiming structural causal intervention.

The C^{2}GES procedure is deterministic once the document, question, role, prediction budget, and scoring weights are fixed. For each sentence, it computes the three scores, forms the weighted sum, sorts all candidates, and returns the top K indices. The experimental implementation is `verification_pilot/scripts/run_c2ges.py`; Algorithm 1 states the executed procedure, and Supplementary Section S3 gives its complexity.

Algorithm 1: C^{2}GES evidence sentence reranking

Require: Document sentences $D = \{s_1, \ldots, s_n\}$, question q, role r, budget K, weights $\theta = (w_q, w_r, w_g)$

for each sentence $s_j \in D$ **do**
    Compute query relevance $Q_j = Q(s_j, q)$
    Compute role compatibility $R_j = R(s_j, r)$
    if $r \in \{\text{propagation/response}, \text{mitigation}\}$ **then**
        Compute chain consistency $G_j = G(s_j, D, r)$
    else
        Set $G_j = 0$
    end if
    Compute $S_j = w_q Q_j + w_r R_j + w_g G_j$
end for
Sort sentence IDs j by descending S_j, using document order for ties
return top-K sentence IDs

The key design choice is the additive score decomposition with a role-selective graph gate. A more complex architecture could learn latent interactions among query, role, and document graph features, but the benchmark size and label provenance favor a transparent reranker whose ablation behavior can be inspected directly. In contrast to prior graph summarization methods that train heterogeneous or multiplex graph neural networks [10,11], C^{2}GES uses graph information as a small consistency signal over candidate sentences. This keeps the method aligned with the central research question: whether causal-role-aware scoring improves sentence-ID evidence recovery for public power-grid maintenance, reliability, disturbance, and lessons-learned report text.

5 Experiments, Results, and Discussion

5.1 Setup, Data, and Baselines

The experiments evaluate whether C^{2}GES improves role-conditioned evidence sentence selection over generic retrieval and extractive summarization baselines. The evaluated benchmark is the local agent-verified candidate dataset at `verification_pilot/agent_audit_40doc/`, containing 40 public NERC reports or report excerpts, 200 causal questions, and 608 evidence sentence IDs. Each document contributes five role-conditioned questions, one for each role: trigger event, root cause, propagation or response, impact, and mitigation. Schema and evidence validation reported 0 errors, so the evaluation uses the provided sentence IDs directly. The labels are agent-rewritten and agent-verified candidate labels, and the experiment measures agreement with those candidate evidence annotations.

The evaluation unit is a document-role question. For each example $(D_i, q_{i,r}, r)$, every method returns $K = 3$ predicted evidence sentence IDs. The primary comparison uses held-out predictions from the Executor artifact, with document-level 5-fold selection for C^{2}GES weight and protocol choices. Fold assignment is deterministic by `sha256(doc_id) modulo 5`, which keeps all questions from the same document in the same fold and prevents document-level leakage across selection and held-out evaluation. The final experiment package was executed once; uncertainty estimates come from document-cluster bootstrap resampling and aggregate variation across evaluation examples, not from multiple random seeds or repeated independent runs. The Executor produced one held-out aggregate per condition, and the tables report the provided aggregate mean and standard-deviation fields from that completed run artifact.

The main conditions are limited to the controlled retrieval and ranking methods used in the benchmark. Positional selection returns the leading sentences and tests whether evidence is concentrated near the start of report excerpts. TF-IDF query retrieval ranks sentences by lexical similarity to the role-conditioned question, while TF-IDF centroid ranks sentences by similarity to the document centroid and therefore represents generic salience. TextRank and LexRank provide graph-centrality summarization baselines, using sentence similarity structure rather than role-specific evidence supervision [9,28]. The causal cue ranker prioritizes sentences with causal trigger language. SBERT query retrieval represents semantic matching with dense sentence embeddings, following the use of BERT-style sentence representations in extractive summarization and retrieval [29,45]. C^{2}GES is the proposed role-aware reranker. A scoped post-freeze supplement additionally evaluates deterministic Okapi BM25 query retrieval on the same sentence pools and role-conditioned questions, using lowercase regex tokenization with $k_1 = 1.5$ and $b = 0.75$, without changing labels, segmentation, C^{2}GES weights, or the primary $K = 3$ protocol. Together, the main comparison set spans eight baseline conditions plus four C^{2}GES variants. Three protocol-matched learned and LLM baselines extend the comparison beyond deterministic methods (Section 5.5), executed with the released harness on the same 200 questions, sentence pools, evaluation metrics, and $K = 3$ budget: a neural cross-encoder reranker (`cross-encoder/ms-marco-MiniLM-L-6-v2`), a BGE reranker

(`BAAI/bge-reranker-base`), and a zero-shot LLM selector (`deepseek-chat`, temperature 0, one API call per question returning exactly three sentence IDs); both neural rerankers ran locally on CPU, and all 200 questions returned valid predictions in every run.

The ablation suite isolates the scoring terms in C^2GES. The query-only ablation retains the query relevance term and removes both role compatibility and chain consistency. The no-role ablation removes $R(s, r)$ while preserving query and chain features, testing whether the causal role signal is responsible for the improvement. The no-graph ablation removes $G(s, D, r)$, testing whether the role-selective causal-chain consistency term changes aggregate evidence recovery. A fixed legacy version of the full C^2GES scorer is retained as a comparison for the revised selection protocol, but the main method remains the full role-selective C^2GES reranker.

The final C^2GES condition uses the selected role-gated-chain configuration recorded in `cv_protocol.json`: $w_q = 0.52$, $w_r = 0.40$, $w_g = 0.08$, with graph active only for propagation/response and mitigation. The candidate grid contains seven lightweight configurations combining additive-chain, role-gated-chain, and fixed legacy scoring families; selection is performed within the document-level 5-fold protocol rather than by hand-tuning on the final aggregate table. Sentence segmentation, feature construction, and baseline candidate pools are shared across conditions so that differences reflect ranking behavior rather than incompatible preprocessing. TF-IDF baselines use the same sentence and question text normalization as the C^2GES lexical query component, and graph-centrality baselines use the same sentence pool as the evidence-ID evaluation. This shared candidate pool is important because the benchmark evaluates sentence IDs directly; a method that selects a near-equivalent text span with a shifted sentence boundary is still penalized under the evidence-ID metric. The complete hyperparameter settings, method abbreviations, and hardware environment are listed in Table S2 of the Supplementary Materials; the selected C^2GES scorer is implemented by the post-freeze role-selective graph-gate runner under `post_freeze_revisions/role_selective_graph_gate/code/main.py`, and placeholder files with null result fields are excluded from the paper's quantitative claims.

5.2 Metrics and Statistical Testing

The evaluation metrics are defined before reporting results because the task is set prediction rather than summary generation. For each example, let $\hat{Y}$ be the predicted top-K sentence-ID set and Y be the candidate evidence set. Evidence precision and recall are

$$P = \frac{|\hat{Y} \cap Y|}{|\hat{Y}|}, \qquad R = \frac{|\hat{Y} \cap Y|}{|Y|}, \tag{6}$$

with higher precision indicating that selected sentences are more frequently annotated as evidence and higher recall indicating that the method recovers more of the annotated evidence. The primary metric, evidence F1, is the harmonic mean

$$F_1 = \frac{2PR}{P + R}, \tag{7}$$

defined as 0 when both P and R are 0. It ranges from 0 to 1, where higher is better, and has no physical unit.

ROUGE-L over selected evidence text is a secondary text-overlap metric. For each example, the predicted sentence texts are concatenated and compared with the concatenated candidate evidence sentence texts using the longest common subsequence formulation of ROUGE-L. Its range is 0 to 1, higher is better, and it measures surface overlap rather than causal adequacy. The main graph and role claims are therefore based on evidence-ID precision, recall, F1, and paired evidence-F1 comparisons; ROUGE-L is used only to check whether better sentence-ID recovery is accompanied by stronger surface overlap. Role coverage is used as a diagnostic to determine whether retrieved evidence spans the five causal roles, while ablation deltas measure the difference in evidence F1 between full C^2GES and each component-removed variant. Qualitative case studies inspect selected and annotated sentence IDs to identify role-mismatch, neighboring-sentence, and implicit-cause errors.

The statistical analysis uses paired comparisons over the same evaluation examples. The predefined primary comparison is full C^2GES against TF-IDF query retrieval on evidence F1, because TF-IDF query retrieval is the strongest lexical retrieval baseline in the main condition set and directly tests whether role-aware reranking adds value beyond question matching. Secondary paired comparisons evaluate C^2GES against SBERT query retrieval, query-only C^2GES, no-role C^2GES, no-graph C^2GES, and the fixed legacy C^2GES scorer. Section 5.4 reports document-cluster paired bootstrap mean differences, 95% confidence intervals, and bootstrap p-values from the Executor artifact. The bootstrap resamples the 40 documents, not the 200 role questions, because questions from the same document share report context, sentence pools, and annotation history. No seed-wise t-tests are reported, and the non-primary p-values are treated as diagnostic evidence rather than as additional primary hypothesis tests.

5.3 Main Quantitative Results

The headline pattern is a consistent relative improvement of role-aware reranking over every generic retrieval and summarization condition. The full C^2GES reranker improves evidence F1 by approximately 41% relative over TF-IDF query retrieval (0.2983 vs. 0.2122), approximately 31% over the strongest supplemental lexical baseline, BM25 query retrieval (0.2273), and approximately 51% over SBERT semantic query retrieval (0.1972), and the paired document-cluster bootstrap analysis in Section 5.4 supports the primary gain over lexical query retrieval at $p < 0.001$. Table 1 reports mean and standard deviation across evaluation examples from the completed Executor run, with BM25 added by the scoped post-freeze supplement; these standard-deviation fields are not confidence intervals. The same pattern appears in precision and recall, where C^2GES improves precision while also retrieving more annotated evidence, so the F1 gain is not driven by a one-sided precision-recall tradeoff. Building on this observation, the selected-evidence ROUGE-L result also increases over TF-IDF, BM25, and SBERT query retrieval, showing that better sentence-ID recovery is accompanied by stronger surface overlap with the annotated evidence text.

Table 1: Performance comparison of different methods on evidence F1, precision, recall, and ROUGE-L (mean ± standard deviation over the 200 evaluation examples; best values per column in bold)

Method	Evidence F1	Precision	Recall	ROUGE-L
Lead sentence selector	0.0545 ± 0.1426	0.0533 ± 0.1392	0.0592 ± 0.1634	0.1822 ± 0.1233
TF-IDF query retrieval	0.2122 ± 0.2293	0.2050 ± 0.2253	0.2471 ± 0.2976	0.3224 ± 0.1931
BM25 query retrieval	0.2273 ± 0.2410	0.2217 ± 0.2410	0.2575 ± 0.2971	0.3247 ± 0.1929
TF-IDF centroid salience	0.0688 ± 0.1556	0.0717 ± 0.1594	0.0717 ± 0.1772	0.2090 ± 0.1360
TextRank with NetworkX	0.0700 ± 0.1549	0.0733 ± 0.1604	0.0725 ± 0.1757	0.2126 ± 0.1297
LexRank	0.0167 ± 0.0946	0.0183 ± 0.1011	0.0154 ± 0.0902	0.1438 ± 0.0810
Causal trigger ranker	0.1071 ± 0.1986	0.1067 ± 0.1993	0.1242 ± 0.2537	0.2302 ± 0.1569
SBERT query retrieval	0.1972 ± 0.2301	0.1883 ± 0.2200	0.2329 ± 0.3037	0.3067 ± 0.1884
Full C^2GES reranker	**0.2983 ± 0.2409**	**0.2950 ± 0.2454**	**0.3325 ± 0.2993**	**0.3732 ± 0.1933**

Absolute scores on this benchmark should be calibrated against the task class rather than against classification-style accuracy. Each question requires selecting exactly three sentence identifiers under five-way causal-role conditioning from the full sentence pool of a long technical report, and the benchmark scores 608 candidate evidence IDs across 40 documents under strict sentence-ID exact matching: a prediction that lands on a semantically adjacent sentence, a shifted segmentation boundary, or a correct-topic but wrong-role sentence receives no credit. Under this protocol, every generic baseline, including a modern dense retriever, remains below 0.23 evidence F1, and unconditioned salience methods stay below 0.11, so an absolute evidence F1 in the 0.3 range is the expected operating regime for this task class rather than a sign of a weak method. This mirrors how strict-match metrics behave in related extraction and selection tasks: in a recent evaluation of open-source LLM agents for SQL generation, strict string-level exact match on the Spider development set stays between 13.44% and 28.14% for the same runs whose execution accuracy exceeds 67% [46], and a study of named-entity extraction from heterogeneous cybersecurity text reports baseline recall of 0.527, with even the best-performing configuration reaching an F1 of only 0.785 [47]. Role-conditioned sentence-ID selection places every question in such a strict-match regime by construction, so the informative quantity is the controlled relative gain over strong retrieval baselines evaluated on identical sentence pools, which is the comparison this benchmark is designed to support.

The Executor artifact does not define easy and hard regimes, so no regime-specific result is inferred; all methods are compared on the same 200 questions. Role-stratified results (Supplementary Section S7) show positive full-C^2GES deltas over TF-IDF and BM25 for every role. Impact has the highest F1 (0.3429) and largest gain over BM25 (+0.1248), whereas propagation/response has the smallest gain (+0.0321), consistent with lexically similar neighboring causal context. The no-role variant falls below BM25 for mitigation and root cause, identifying role compatibility rather than generic relevance as the main source of improvement.

Cross-document heterogeneity is substantial and is reported as an honest robustness observation rather than smoothed away. The standard deviations in Table 1 are computed over the 200 question-level scores and are of the same order as the means (about 0.24 against a mean of 0.2983 for the full reranker), which is expected for a strict top-K set-overlap metric scored per question. Aggregating to the document level, per-document mean evidence F1 for full C^2GES ranges from 0.0800 to 0.5219 across the 40 documents with no document at zero; the full per-document dispersion details for C^2GES and the query baselines are reported in the Supplementary Materials (Section S2). Because documents, not questions, are the natural unit of this heterogeneity, all interval estimates in this paper use document-cluster bootstrap resampling rather than question-level resampling.

5.4 Paired Comparisons and Ablations

The paired comparisons confirm that C^2GES's advantage is strongest against query-only retrieval and role-removed variants. Table 2 summarizes document-cluster paired bootstrap comparisons for evidence F1. The difference against TF-IDF query retrieval supports the pre-specified primary comparison, while the remaining comparisons are diagnostic. The comparison against SBERT query retrieval supports the same qualitative conclusion for semantic matching. The comparison against the no-role ablation identifies role compatibility as a major source of the aggregate improvement. The comparison against the no-graph ablation is much smaller, but the role-selective gate produces a positive auxiliary graph/chain gain under document-cluster bootstrap. For transparency, the role-selective gate itself originated in a post-freeze revision: an earlier frozen configuration applied the graph term to all five roles and showed no reliable graph benefit, and the gated variant reported here was selected afterwards within the document-level 5-fold candidate protocol, so its $p = 0.0254$ should be read as diagnostic support for a small auxiliary effect rather than as a confirmatory pre-registered test.

A bounded K-sensitivity check for the fixed final scorer and the key retrieval baselines is reported in full in the Supplementary Materials (Table S1), together with the per-budget paired bootstrap comparisons.

Table 2: Document-cluster paired bootstrap comparisons for evidence F1

Comparison	Mean F1 Difference	Document-Cluster 95% CI	p-Value	Interpretation
Full C^{2}GES vs. TF-IDF query retrieval	0.0861	[0.0581, 0.1145]	<0.001	Primary gain over lexical query retrieval
Full C^{2}GES vs. BM25 query retrieval	0.0710	[0.0423, 0.1000]	<0.001	Supplemental gain over BM25 lexical retrieval
Full C^{2}GES vs. SBERT query retrieval	0.1010	[0.0629, 0.1388]	<0.001	Diagnostic gain over semantic query retrieval
Full C^{2}GES vs. C^{2}GES query-only	0.0831	[0.0541, 0.1120]	<0.001	Role and structure terms improve query-only ranking
Full C^{2}GES vs. C^{2}GES without role compatibility	0.0688	[0.0416, 0.0972]	<0.001	Role compatibility is important
Full C^{2}GES vs. C^{2}GES without graph consistency	0.0060	[0.0014, 0.0119]	0.0254	Small role-selective graph gain
Full C^{2}GES vs. fixed legacy C^{2}GES	0.0403	[0.0178, 0.0621]	<0.001	Revised protocol improves over fixed legacy scoring

The primary benchmark budget remains $K = 3$, and the additional $K = 1$ and $K = 5$ evaluations are not used to select a new operating point. In summary, the C^{2}GES advantage is largest and statistically supported at $K = 3$, where the document-cluster bootstrap gain over BM25 is +0.0710 evidence F1 with 95% CI $[0.0423, 0.1000]$; at $K = 5$, C^{2}GES remains ahead of BM25 by +0.0643 with 95% CI $[0.0331, 0.0961]$; and at $K = 1$, C^{2}GES has the highest aggregate F1, but its paired gains over TF-IDF and BM25 have confidence intervals that cross zero, so the stricter one-sentence setting should be interpreted as suggestive rather than conclusive. The supplemental comparisons come from a single run artifact and are not corrected for multiple comparisons; only the TF-IDF comparison at $K = 3$ retains pre-specified primary status.

The ablations separate role awareness from graph enhancement. Query-only C^{2}GES reaches 0.2152 evidence F1, no-role 0.2295, and no-graph 0.2923, versus 0.2983 for the full method. Thus role compatibility is the dominant mechanism, while graph consistency adds 0.0060 overall; its role-stratified gains are confined to propagation/response (+0.0083) and mitigation (+0.0214), as detailed in Supplementary Section S7. The evidence therefore supports a role-aware reranker with a lightweight structural auxiliary term, not graph-driven causal inference.

5.5 Learned and LLM Baselines

Table 3 reports the three protocol-matched learned and LLM baselines alongside the key existing conditions, with paired document-cluster bootstrap differences against the full C^{2}GES reranker (10000 samples; delta defined as baseline minus C^{2}GES, so negative values favor C^{2}GES); the full paired-comparison detail and the measured LLM run cost are given in the Supplementary Materials (Section S6).

Within the transparent CPU-deployable reranker class, full C^{2}GES has the best point estimate: its advantage over BGE is borderline significant ($p = 0.0448$), whereas its advantage over the cross-encoder is not significant ($p = 0.246$). The zero-shot LLM is substantially stronger (+0.2905 F1, $p < 0.001$), but the agent-generated label provenance makes 0.5887 an upper reference pending human-gold evaluation. It also requires 200 remote calls and offers no per-term decomposition, while C^{2}GES is deterministic and exposes

Table 3: Learned and LLM baselines under the paper's protocol ($K = 3$, 200 questions, same metrics), with paired comparison against full C^{2}GES (delta = baseline − C^{2}GES)

Method	F1	Precision	Recall	ROUGE-L	Difference vs. Full C^{2}GES [95% CI]	p
SBERT query retrieval	0.1972	0.1883	0.2329	0.3067	−0.1010 [−0.1388, −0.0629]	<0.001
TF-IDF query retrieval	0.2122	0.2050	0.2471	0.3224	−0.0861 [−0.1145, −0.0581]	<0.001
BM25 query retrieval	0.2273	0.2217	0.2575	0.3247	−0.0710 [−0.1000, −0.0423]	<0.001
BGE reranker (bge-reranker-base)	0.2604	0.2517	0.3013	0.3490	−0.0379 [−0.0739, −0.0010]	0.0448
Cross-encoder (ms-marco-MiniLM)	0.2787	0.2717	0.3188	0.3667	−0.0196 [−0.0539, +0.0140]	0.246
Full C^{2}GES reranker	0.2983	0.2950	0.3325	0.3732	—	—
LLM zero-shot (deepseek-chat)	**0.5887**	**0.5950**	**0.6342**	**0.5942**	+0.2905 [+0.2533, +0.3255]	<0.001

three published score components. Supplementary Section S6 reports the complete intervals, costs, and deployment qualifications.

5.6 Qualitative Error Analysis

Held-out predictions illustrate both the benefit and the remaining error modes. For the solar-PV impact case, full C^{2}GES retrieves two of three labeled sentences (F1 0.667 versus 0.0 for TF-IDF) by favoring measured output and net-load consequences. For the Odessa root-cause case, it retrieves topically related protection and generation-loss sentences but misses the failed surge arrester (F1 0.0 versus 0.4), a neighboring-context role error. A further trigger-event failure is documented in Supplementary Section S5. These cases motivate better sentence-boundary handling and local context aggregation, while supporting graph consistency only as an auxiliary signal for structurally contextual roles.

5.7 Discussion

Role-aware reranking outperforms generic relevance retrieval because a sentence can concern the same disturbance yet answer the wrong causal role. This extends query-conditioned selection [5,6] to sentence-ID evidence retrieval [34,36] in a grid setting where initiating events, propagation, impacts, and corrective actions are distinct analytical needs [1,40].

The ablations locate the empirical contribution: role compatibility matters substantially more than the lightweight graph term. Rich graph summarizers can model broader relations [10,11], but the present chain feature yields only a small role-selective gain. Consistent with cautions in causal NLP [17], this is a diagnostic component-removal result, not counterfactual causal inference; future effort should prioritize role modeling unless stronger edge induction yields larger gains.

Protocol-matched learned and LLM baselines clarify the tradeoff [37–39]: the zero-shot LLM is much more accurate, whereas C^{2}GES provides deterministic, published, per-term score decomposition for audit. It is therefore an analyst-facing ranking aid, not an expert replacement or operational event-reconstruction system; selected sentence IDs should be reviewed and corrected before supporting regulatory or corrective-action claims.

6 Conclusions, Limitations, and Future Work

C^{2}GES frames power-grid maintenance and reliability report analysis as causal-role evidence sentence selection and contributes a scoped candidate-benchmark-driven reranker for public NERC reports or report excerpts. The main result is that role-aware reranking outperforms generic lexical and semantic query retrieval, with relative evidence-F1 gains of approximately 41% over TF-IDF, 31% over BM25, and 51% over SBERT query retrieval, and with ablations showing that role compatibility is the primary source of improvement while role-selective graph consistency adds a small auxiliary gain for propagation/response and mitigation. Protocol-matched learned and LLM baselines position the method honestly: full C^2GES leads the transparent, CPU-deployable reranker class (borderline significantly ahead of a BGE reranker, statistically indistinguishable from a cross-encoder), while zero-shot LLM selection roughly doubles evidence F1 as an upper reference that offers no per-term score decomposition and requires a remote API.

The conclusions are bounded by several limitations. First, the study is scoped to a 40-document local benchmark of public NERC reports and report excerpts, some of which are event-focused excerpts inside larger reliability or lessons-learned reports, so the available context sometimes omits wider operational details that an expert analyst might use, and the findings concern evidence selection on this benchmark rather than broad generalization across all grid maintenance documentation. Second, the 608 evidence labels are agent-rewritten, agent-verified candidate labels rather than human-gold annotations, no expert adjudication or inter-annotator agreement estimate is available for this pilot, verification depth is uneven across documents (of the 15 later-added documents, 6 passed cleanly, 8 required minor repairs, 1 required answer narrowing, and one document remains flagged for low evidence diversity), and the role cue lexicons were curated during method development on the same corpus without a held-out protocol protecting cue authorship, so a label–cue circularity component in the role-compatibility gain cannot be excluded (Section 4.4); the planned human-gold subset study with dual annotation and Cohen's κ is the intended verification. Third, the final experiment package was executed once, and uncertainty estimates come from document-cluster bootstrap resampling rather than repeated independent runs. Fourth, C^2GES does not estimate structural causal effects, simulate power-system counterfactuals, or learn a full graph neural representation of report structure, and the learned and LLM baseline coverage of Section 5.5, while now present, is bounded: RM3, SPLADE, ColBERT, monoT5, learned-to-rank, and fine-tuned domain-adapted rerankers [37,38] are not evaluated, and the LLM baseline is a single model evaluated zero-shot with one fixed prompt, subject to the LLM–LLM label-provenance caveat of Section 4.4. Fifth, the evaluation uses $K = 3$ as the primary budget with only a bounded $K \in \{1, 3, 5\}$ sensitivity check, segmentation and graph-window sensitivity are not reported, and the strict top-K sentence-ID metric counts near-miss sentences, shifted segmentation, and partially correct neighboring evidence as errors even when they may still be useful to a reviewer.

Future work should add expert-adjudicated labels with inter-annotator agreement (which would also arbitrate the LLM baseline's advantage under independent labels), broader learned baselines including fine-tuned domain rerankers and multi-model LLM selection with prompt-sensitivity analysis, predeclared segmentation sensitivity studies, stronger role-transition modeling, and regime-level error analysis. A useful next step is an analyst-in-the-loop interface that turns selected evidence IDs into reviewable causal summaries without making the generated summary the evaluated object.

Acknowledgement: During the preparation of this work, the authors used large-language-model-based agent tooling in order to construct and verify the benchmark labels: the evidence labels evaluated in this study are agent-rewritten and agent-verified candidate labels produced with such tooling, as disclosed in Section 3, and all quantitative claims are scoped to agreement with these candidate labels. The authors also used large-language-model-based tools in order to assist with drafting and editing the manuscript text. After using these tools and services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Availability of Data and Materials: The data that support the findings of this study are openly available in the c2ges repository at https://github.com/gaoxingkele/c2ges (release v0.2.0), including the benchmark dataset workspace (40 public NERC reports or report excerpts segmented into 2940 sentences, 200 role-conditioned questions, and 608 candidate evidence sentence IDs, with the corpus manifest mapping each document to its official public source URL), the complete original Executor run artifacts (summary, per-question details, cross-validation protocol record, and held-out predictions, independently verified against every reported table), the evidence supplement supporting the reported tables (seven-condition K-sensitivity run, paired bootstrap comparisons, and role-stratified metrics), the learned and LLM baseline run outputs of Section 5.5, the analysis and verification code, and the baseline harness. The underlying source documents are public NERC reliability, disturbance, and lessons-learned materials. A detailed mapping from each reported table to its evidencing artifact is provided in the Supplementary Materials (Section S4).

Ethics Approval: Not applicable. This study did not involve human or animal subjects; it analyzes publicly available NERC reliability reports.

Conflicts of Interest: The authors declare no conflicts of interest.

Supplementary Materials: The supplementary material is available online at https://www.techscience.com/doi/10.32604/cmc.2026.000000/s1, Table S1: supplemental K-sensitivity evidence F1 with per-budget paired bootstrap comparisons; Table S2: hyperparameter settings, method abbreviations, and hardware environment; Section S2: per-document dispersion details; Section S3: computational complexity analysis; Section S4: artifact-to-table evidence map; Section S5: additional qualitative case analysis; Section S6: paired-comparison detail and run cost for the learned and LLM baselines; Section S7: complete deterministic scoring specification and role-stratified results.

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